

# SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 04.3

Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB.

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## Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

## Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

## MATLAB Program

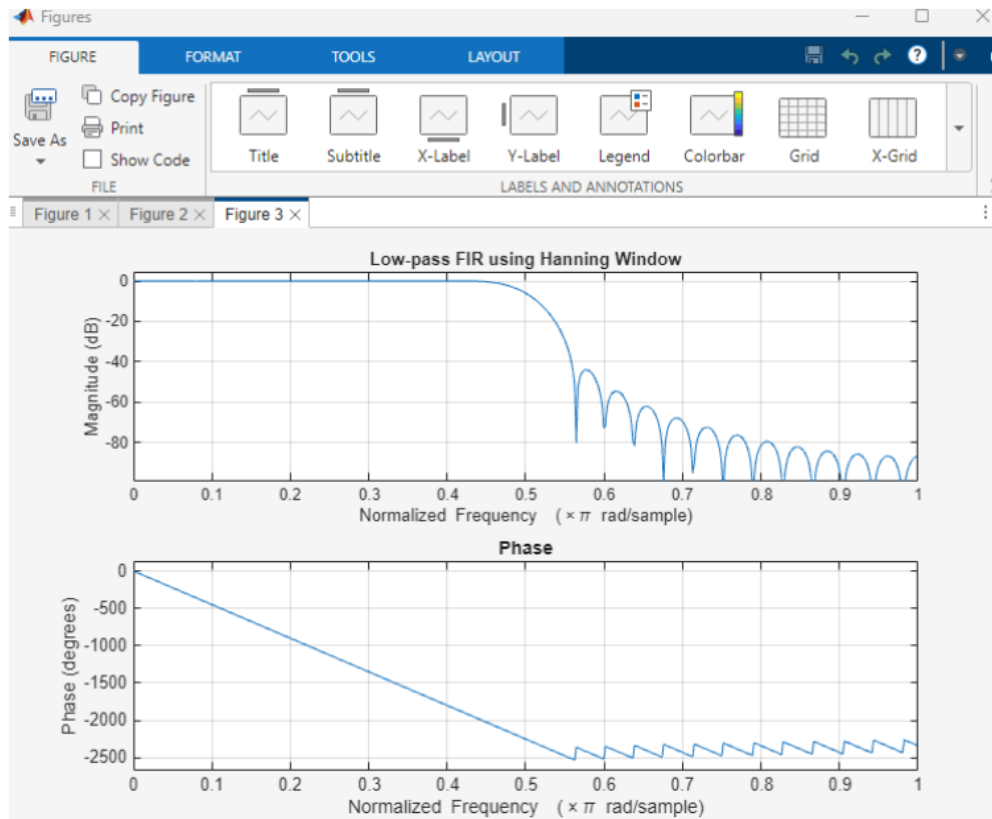
```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000;      % Passband cutoff (Hz)
fs = 4000;      % Sampling frequency (Hz)
N = 50;         % Filter order (even number preferred)
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
%% --- FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);
%% --- Plot Frequency Response ---
figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');
```

## Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
%% --- FIR Low Pass using Hanning Window ---  
  
b_hann = fir1(N, wc, 'low', hanning(N+1));  
disp('--- Hanning Window Coefficients ---');  
disp(b_hann);  
  
%% --- Plot Frequency Responses ---  
  
figure;  
freqz(b_hann,1);  
  
title('Low-pass FIR using Hanning Window');  
  
OUTPUT;  
  
--- Hanning Window Coefficients ---  
0.0000  
-0.0000  
-0.0004  
0.0000  
0.0013  
-0.0000  
-0.0028  
0.0000  
0.0050  
-0.0000  
-0.0081  
0.0000  
0.0122  
-0.0000  
-0.0179  
0.0000  
0.0259  
-0.0000  
-0.0378  
0.0000  
0.0580  
-0.0000  
-0.1027  
0.0000  
0.3172  
0.5000  
0.3172  
0.0000  
-0.1027
```

-0.0000  
0.0580  
0.0000  
-0.0378  
-0.0000  
0.0259  
0.0000  
-0.0179  
-0.0000  
0.0122  
0.0000  
-0.0081  
-0.0000  
0.0050  
0.0000  
-0.0028  
-0.0000  
0.0013  
0.0000  
-0.0004  
-0.0000  
0.0000



## Result

1. The FIR Low-Pass Filter was successfully designed using the Hanning Window technique for the specified filter parameters.
2. The FIR filter coefficients were generated and implemented successfully in MATLAB.
3. The magnitude and phase responses of the designed filter were obtained and analyzed.
4. The filter exhibited the desired low-pass characteristics with reduced sidelobes and improved stopband attenuation compared to the Rectangular Window method.

## Practical Question 1

Design an FIR Low-Pass Filter using the **Hanning (Hann) Window** technique in MATLAB for a sampling frequency of **8000 Hz**, cutoff frequency of **1500 Hz**, and filter order of **50**.

### Tasks:

1. Calculate the normalized cutoff frequency.
2. Generate the FIR filter coefficients using the `fir1()` function with a Hanning Window.

3. Plot the magnitude response and phase response of the designed filter.
4. Analyze the passband, transition band, and stopband characteristics.

| Aim<br>(10) | Program<br>(20) | Procedure/Calculations<br>(20) | Results<br>(20) | Record<br>(10) | Viva<br>(20) | Total<br>(100) |
|-------------|-----------------|--------------------------------|-----------------|----------------|--------------|----------------|
|             |                 |                                |                 |                |              |                |

### Practical Question 2

Design and implement an FIR Low-Pass Filter using the **Hanning (Hann) Window** technique in MATLAB for a sampling frequency of **12000 Hz**, cutoff frequency of **2000 Hz**, and filter order of **60**.

#### Tasks:

1. Design the FIR Low-Pass Filter using the specified parameters and Hanning Window.
2. Generate and display the FIR filter coefficients.
3. Obtain the magnitude response and phase response using the `freqz()` function.
4. Determine the cutoff frequency and stopband attenuation from the response plots.

| Aim<br>(10) | Program<br>(20) | Procedure/Calculations<br>(20) | Results<br>(20) | Record<br>(10) | Viva<br>(20) | Total<br>(100) |
|-------------|-----------------|--------------------------------|-----------------|----------------|--------------|----------------|
|             |                 |                                |                 |                |              |                |

## Result

To design and analyze an FIR low-pass filter using a Hanning window and examine its passband, transition band, and stopband.

## Student Details

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